

SENSORS IN PCBA

Assembly, Types & Integration Guide

Temperature • Pressure • Humidity • Proximity • Accelerometer • Gas
MEMS • IoT • Automotive • Placement • Soldering • Calibration

PCBA Manufacturing & Sensor Integration Standards

IPC/J-STD-001 | IPC-A-610 | IPC-7093 | Automotive AEC-Q100

1. SENSORS IN PCBA OVERVIEW

Sensors are critical components in modern PCBA assemblies that detect and measure physical, chemical, or biological properties and convert them into electrical signals. From automotive engine control to smartphone motion sensing, sensors enable intelligent functionality in nearly every electronic device.

Why Sensors are Critical in PCBA:

- **Environmental awareness:** Temperature, humidity, pressure monitoring
- **Safety systems:** Proximity, collision, airbag deployment sensors
- **Motion control:** Accelerometers, gyroscopes for navigation and gaming
- **Health monitoring:** Heart rate, blood oxygen, biometric sensors
- **Industrial automation:** Position, force, vibration sensors
- **IoT connectivity:** Smart home, smart city sensor networks

Sensor Assembly Challenges:

- **Precision placement:** MEMS sensors require exact orientation
- **ESD sensitivity:** Many sensors are damaged by electrostatic discharge
- **Moisture sensitivity:** MEMS devices require controlled storage
- **Calibration:** Post-assembly calibration is often required
- **Shielding:** Magnetic and EMI sensors need isolation from noise sources
- **Conformal coating:** Environmental sensors must remain exposed

■ **In-House Note:** Sensor assembly requires special handling compared to standard passives and ICs. Many sensors (especially MEMS) have MSL (Moisture Sensitivity Level) ratings of 3 or higher, requiring bake-out before reflow and controlled storage after opening.

2. SENSOR TYPES IN PCBA ASSEMBLY

2.1 Temperature Sensors

Temperature sensors measure thermal conditions for system monitoring, thermal management, and safety shutdown.

Types:

- **NTC Thermistors:** Negative Temperature Coefficient resistors; resistance decreases with temperature. 0603/0402 SMD packages. Simple, low cost.
- **PTC Thermistors:** Positive Temperature Coefficient; used for overcurrent protection. SMD disc or chip packages.
- **RTD (Resistance Temperature Detector):** Platinum thin-film (Pt100, Pt1000). High accuracy, linear response. SMD or wired packages.
- **Digital Temperature Sensors:** IC-based (LM75, TMP102, SHT series). I2C/SPI output. QFN, SOT-23, WLCSP packages.
- **Infrared (IR) Sensors:** Non-contact temperature measurement. MLX90614, TMP006. SMD packages with optical window.

Assembly Notes:

- Place away from heat-generating components (CPUs, power regulators)
- Ensure good thermal coupling to measured surface
- Avoid conformal coating over sensing element (if exposed)
- NTC thermistors: Standard reflow; no special handling

2.2 Pressure Sensors

Pressure sensors measure gas or liquid pressure for automotive, medical, industrial, and consumer applications.

Types:

- **MEMS Piezoresistive:** Silicon diaphragm with piezoresistors. MPXH series (NXP), BMP280 (Bosch). LGA, QFN packages with pressure port.
- **MEMS Capacitive:** Diaphragm changes capacitance. Higher sensitivity than piezoresistive. Used in high-precision applications.
- **Absolute Pressure:** Measures relative to vacuum. Used for altitude, barometric sensing.
- **Gauge Pressure:** Measures relative to atmospheric pressure. Used for tire pressure, fluid monitoring.
- **Differential Pressure:** Measures difference between two ports. Used in HVAC, medical ventilators.

Assembly Notes:

- **Pressure port orientation:** Must face correct direction; verify mechanical drawing
- **Seal integrity:** Gasket or O-ring must be properly seated
- **Reflow:** Standard SAC305 profile; avoid pressure on sensor body
- **Cleaning:** Do NOT use high-pressure spray directly on pressure port

2.3 Humidity Sensors

Humidity sensors measure relative humidity (RH) for environmental control, weather stations, and industrial processes.

Types:

- **Capacitive RH Sensors:** Polymer dielectric changes capacitance with humidity. SHT series (Sensirion), HIH series (Honeywell). SMD packages with exposed sensing element.
- **Resistive RH Sensors:** Conductive polymer changes resistance. Lower accuracy than capacitive.
- **Thermal Conductivity:** Measures absolute humidity. Used in industrial applications.

Assembly Notes:

- **Exposed element:** Sensing window must remain uncovered by solder mask or coating
- **Handling:** Fingerprints and contaminants affect accuracy — use gloves
- **Storage:** Keep in original packaging until assembly; avoid high humidity exposure
- **Reflow:** Standard profile; avoid flux residue on sensing element

2.4 Proximity & Optical Sensors

Proximity sensors detect the presence or absence of objects without physical contact. Optical sensors detect light intensity, color, or specific wavelengths.

Types:

- **IR Proximity:** Emit IR LED, measure reflected light. VCNL series (Vishay). SMD packages with integrated emitter/detector.
- **Time-of-Flight (ToF):** Measure light travel time. VL53L series (STM). Very compact SMD.
- **Ambient Light Sensors:** Measure visible light for auto-brightness. ALS series. Small SMD or chip-scale.
- **Color Sensors:** RGB or multi-spectral detection. TCS series (AMS). SMD packages.
- **PIR (Passive Infrared):** Detect motion via heat radiation. Through-hole or SMD modules.

Assembly Notes:

- **Optical window:** Must remain clean and unobstructed
- **Alignment:** Emitter and detector must face correct direction
- **Reflow:** Standard profile; avoid flux on optical surfaces
- **ESD:** Photodiodes are ESD sensitive — handle carefully

2.5 Motion & Position Sensors

Motion sensors detect acceleration, rotation, and orientation for navigation, gaming, image stabilization, and vehicle dynamics.

Types:

- **Accelerometers:** MEMS cantilever measures linear acceleration. ADXL series (Analog Devices), LIS series (STM). LGA, QFN, WLCSP packages.
- **Gyroscopes (MEMS):** Measure angular velocity. L3G series (STM), ITG series (InvenSense). Similar packages to accelerometers.
- **IMU (Inertial Measurement Unit):** Combined accelerometer + gyroscope + magnetometer. MPU series (InvenSense/TDK). Compact QFN or BGA.
- **Magnetometers:** Measure magnetic field direction. HMC series (Honeywell), AK series (AKM). Small SMD packages.
- **Encoders:** Optical or magnetic position sensing. Through-hole or connector-mounted modules.

Assembly Notes:

- **Orientation:** Package marking indicates axis direction — verify placement angle
- **Mounting:** Rigid mounting required — flexible PCB causes measurement errors
- **Reflow:** Standard SAC305; avoid ultrasonic cleaning (damages MEMS)
- **Calibration:** Post-assembly 6-point calibration often required

2.6 Gas & Chemical Sensors

Gas sensors detect specific gases (CO, CO₂, VOCs, O₂) for safety, environmental monitoring, and industrial control.

Types:

- **Electrochemical:** Gas reacts with electrolyte; measures current. CO, H₂S sensors. Through-hole or connector modules.
- **Metal Oxide Semiconductor (MOS):** Resistance changes with gas concentration. MiCS series (SGX). SMD packages with exposed sensing layer.
- **Infrared (NDIR):** Non-dispersive infrared absorption. CO₂ sensors. Larger modules with optical path.
- **Catalytic Bead:** Combustible gas detection. Industrial safety applications.

Assembly Notes:

- **Ventilation:** Sensing element must have air access — do not encapsulate
- **Heating element:** MOS sensors have integrated heater — verify power supply
- **Reflow:** Standard profile; avoid high-temp exposure beyond specification
- **Burn-in:** MOS sensors require 24–48 hour burn-in for stable baseline

3. MEMS SENSORS — SPECIAL CONSIDERATIONS

3.1 What is MEMS?

MEMS (Micro-Electro-Mechanical Systems) are miniaturized mechanical and electro-mechanical elements manufactured using semiconductor fabrication techniques. MEMS sensors combine mechanical structures (cantilevers, diaphragms, proof masses) with integrated electronics on a single silicon chip.

MEMS Manufacturing:

- **Wafer fabrication:** Standard CMOS processes with deep reactive ion etching (DRIE)
- **Packaging:** Vacuum-sealed or hermetically sealed cavities for moving parts
- **Wire bonding:** Gold or aluminum wire bonds connect die to package leads
- **Molding:** Epoxy molding compound protects wire bonds and die

3.2 MEMS Package Types

Package	Size	I/O	Applications	Assembly Notes
LGA	2×2 to 5×5 mm	8–16	Accelerometer, pressure	No leads; solder paste on pads
QFN	3×3 to 10×10 mm	16–64	Gyroscope, IMU	Exposed pad for grounding/thermal
WLCSP	1.5×1.5 to 3×3 mm	4–25	Compact accelerometer	Die-size; underfill recommended
BGA	4×4 to 15×15 mm	48–400	Complex IMU, combo sensors	Standard BGA assembly rules
SIP (System-in-Package)	5×5 to 15×15 mm	Multiple die	Multi-sensor modules	Multiple die + passive integration

Table 1: MEMS Sensor Package Types

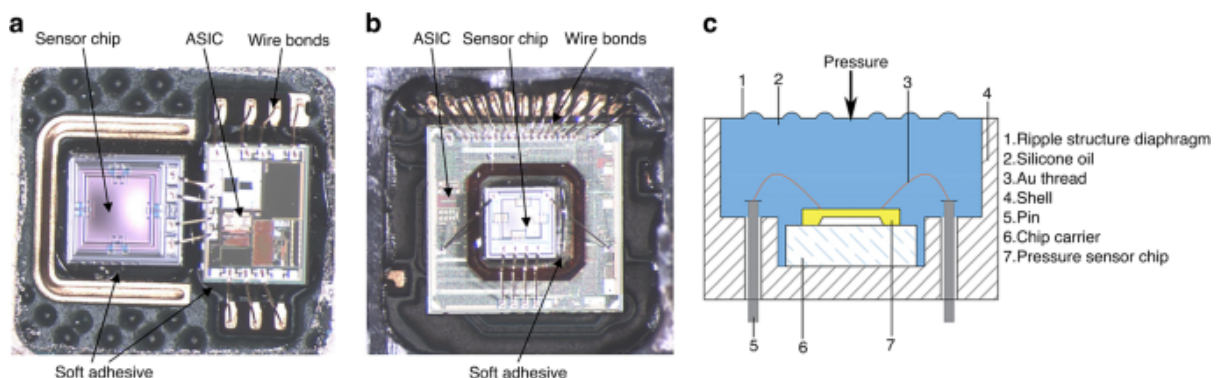


Figure 1: MEMS Pressure Sensor Cross-Section — Sensor Chip, ASIC, Wire Bonds, Soft Adhesive

3.3 MEMS Handling Precautions

MEMS sensors contain microscopic moving parts that are easily damaged by improper handling.

- **NO Ultrasonic Cleaning:** Ultrasonic cavitation destroys MEMS structures. Use spray-in-air or DI water only.
- **NO Drop/Shock:** Mechanical shock can break cantilevers or proof masses. Handle with care.
- **ESD Protection:** MEMS ASICs are ESD sensitive. Handle in controlled environments.
- **Moisture Sensitivity:** Most MEMS are MSL 3 or higher. Bake 125°C/24 hours before reflow if exposed.
- **Vacuum Integrity:** Do NOT puncture package cavity. Vacuum loss degrades performance.
- **Orientation:** Place according to datasheet marking. Wrong orientation = wrong axis measurement.

■■ **CRITICAL:** Never subject MEMS sensors to ultrasonic cleaning, drop testing, or excessive mechanical stress. Even minor damage to the internal mechanical structure causes permanent performance degradation that may not be detectable during basic electrical test.

4. SENSOR PLACEMENT & PCB LAYOUT

4.1 General Placement Rules

Sensor placement affects measurement accuracy, thermal performance, and signal integrity.

- **Temperature sensors:** Place away from heat sources (CPUs, regulators, power FETs). Minimum 10 mm separation.
- **Pressure sensors:** Orient pressure port to face measurement medium. Ensure mechanical clearance for tubing.
- **Humidity sensors:** Place in airflow path, not enclosed corners. Avoid direct exposure to condensation.
- **Optical sensors:** Face toward intended detection area. Avoid shadowing by tall components.
- **Motion sensors:** Mount at board center of rotation for gyroscope accuracy. Rigid mounting essential.
- **Magnetic sensors:** Isolate from ferrite inductors, transformers, and current-carrying traces.
- **Gas sensors:** Ensure air access. Do not place under shields or conformal coating.

4.2 Analog vs. Digital Sensor Placement

Analog sensors require more careful layout than digital sensors due to noise sensitivity.

Analog Sensors (Thermistors, RTDs, analog accelerometers):

- Keep traces short and shielded
- Use dedicated analog ground plane
- Place away from switching regulators and digital clocks
- Use differential routing for bridge sensors

Digital Sensors (I2C, SPI, UART):

- Standard digital routing rules apply
- I2C pull-up resistors close to sensor
- SPI traces matched length for high-speed clocks

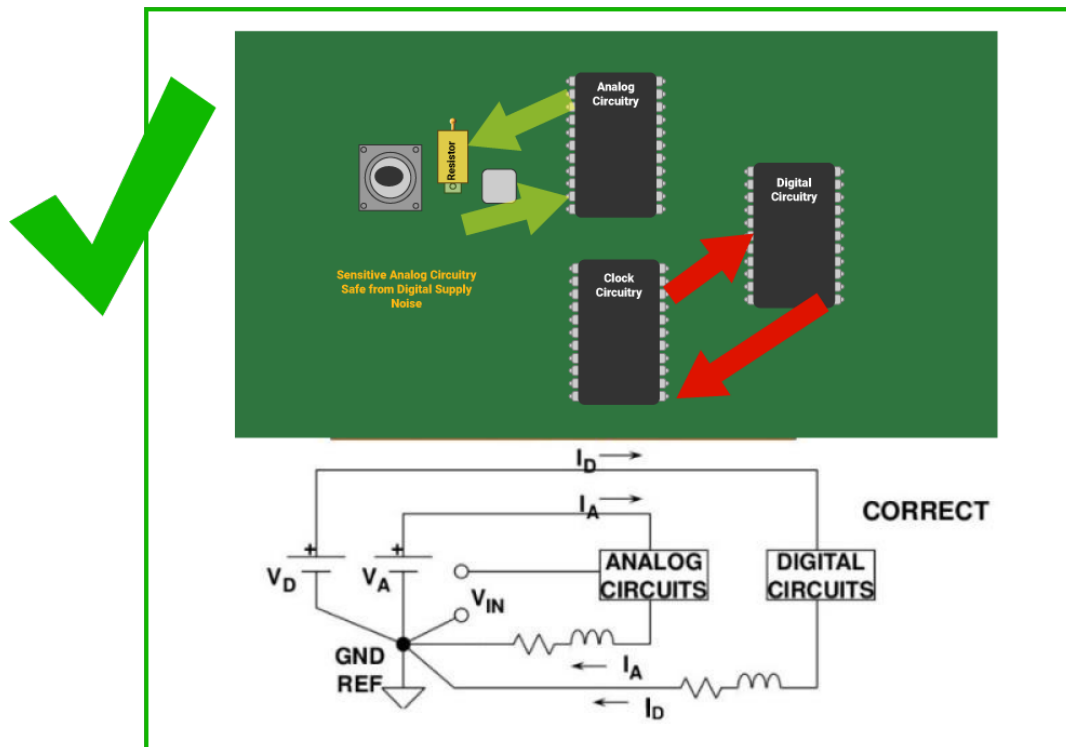


Figure 2: Sensor Placement Guidelines — Analog vs. Digital Circuitry Separation

4.3 Grounding and Shielding

Proper grounding is critical for sensor accuracy, especially for magnetic and analog sensors.

- **Star grounding:** Single point connection between analog and digital grounds
- **Ground planes:** Solid ground plane under sensor; no splits or slots
- **Faraday cages:** Copper shield cans for EMI-sensitive sensors (GPS, magnetometers)
- **Guard rings:** Surround high-impedance analog inputs with grounded tracks
- **Decoupling:** 100 nF ceramic + 10 μ F tantalum close to sensor power pins

5. SENSOR SOLDERING & REFLOW

5.1 Standard Reflow Profile

Most sensors use standard SAC305 reflow profiles. However, some sensors have special temperature limits.

Standard SAC305 Profile (for most sensors):

- **Preheat:** 25 → 150°C at 1.5–2.5°C/sec
- **Soak:** 150 → 180°C for 60–90 seconds
- **Peak:** 245–250°C
- **TAL:** 60–90 seconds above 217°C
- **Cooling:** 2–4°C/sec

5.2 Temperature-Sensitive Sensors

Some sensors have maximum temperature limits below standard reflow peak.

- **Polymer humidity sensors:** Max 240°C peak — use lower profile
- **Electrochemical gas sensors:** Max 230°C — hand solder or conductive epoxy
- **Optical sensors with plastic lenses:** Max 245°C — verify datasheet
- **MEMS with organic overmold:** Standard profile acceptable; verify MSL rating

5.3 Hand Soldering Guidelines

Some sensors (modules, through-hole, or temperature-sensitive) require hand soldering.

- **Temperature:** 315–350°C iron tip
- **Dwell time:** 2–5 seconds per pin
- **Heat sink:** Use heatsink clip on sensor body if sensitive
- **ESD:** Grounded iron and wrist strap mandatory
- **Flux:** No-clean RMA or ORH1; clean residue if water-soluble

5.4 Post-Reflow Cleaning

Cleaning sensors requires special care to avoid damage.

- **MEMS sensors:** NO ultrasonic cleaning — spray-in-air or manual wipe only
- **Optical sensors:** Do not allow cleaning solvent on lens — mask before cleaning
- **Pressure sensors:** Do not force solvent into pressure port
- **Humidity sensors:** Ensure sensing element is completely dry before power-on
- **Gas sensors:** Allow 24-hour dry-out after any cleaning before calibration

■ **In-House Best Practice:** For MEMS sensors, add a note on the work order: 'NO ULTRASONIC CLEANING — Spray-in-air or manual wipe only.' This prevents accidental damage during the cleaning process.

6. SENSOR CALIBRATION & TESTING

6.1 Why Calibration is Needed

Sensors have manufacturing tolerances that cause offset and gain errors. Calibration corrects these errors to meet accuracy specifications.

Calibration Types:

- **Factory calibration:** Performed by sensor manufacturer; stored in EEPROM
- **System calibration:** Performed after PCBA assembly; compensates for PCB effects
- **Field calibration:** Periodic recalibration during product lifetime

6.2 Common Calibration Methods

Temperature Sensors:

- **Ice point (0°C):** Reference junction in ice bath
- **Boiling point (100°C):** Steam reference (adjusted for altitude)
- **Dry block calibrator:** Precision heated block with known temperature
- **Two-point calibration:** Measure at 25°C and 75°C; calculate offset and gain

Pressure Sensors:

- **Zero calibration:** Apply vacuum or atmospheric pressure; set zero output
- **Span calibration:** Apply known pressure (e.g., 100 kPa); set full-scale output
- **Dead weight tester:** Precision mechanical pressure standard

Accelerometers & Gyroscopes:

- **6-point calibration:** Orient sensor at +1g and -1g on each axis
- **Zero-rate calibration:** Measure output at rest; subtract offset
- **Scale factor:** Known rotation rate (e.g., 90°/sec) for gyroscope

Humidity Sensors:

- **Salt solution method:** Saturated salt solutions at known RH (e.g., NaCl = 75% RH)
- **Two-point:** 11% RH (LiCl) and 75% RH (NaCl)
- **Reference hygrometer:** Precision instrument for comparison

6.3 Production Testing

Production tests verify sensor functionality without full calibration.

- **Continuity:** Verify all pins connected (no opens or shorts)
- **Power consumption:** Verify current draw within specification
- **Basic response:** Apply stimulus and verify output changes (e.g., warm finger on temp sensor)
- **Communication:** Verify I2C/SPI/UART responds to read commands

- **Self-test:** Use built-in self-test (BIST) if available

7. AUTOMOTIVE SENSOR REQUIREMENTS

7.1 AEC-Q100 Qualification

AEC-Q100 is the automotive electronics council standard for integrated circuit qualification. Sensors used in automotive applications (Tesla, Zoox) must meet AEC-Q100 Grade 0, 1, 2, or 3 depending on temperature range.

Grade	Temperature Range	Use Case	Examples
Grade 0	-40°C to +150°C	Under-hood, powertrain	Engine temp, oil pressure
Grade 1	-40°C to +125°C	Cabin, body electronics	HVAC, seat occupancy
Grade 2	-40°C to +105°C	Infotainment, lighting	Ambient light, touchscreen
Grade 3	-40°C to +85°C	Non-critical	Clock, convenience

Table 2: AEC-Q100 Temperature Grades

7.2 Automotive Sensor Testing

Automotive sensors require additional testing beyond standard PCBA functional test.

- **HTOL (High Temperature Operating Life):** 1000 hours at maximum temperature
- **Temperature cycling:** -40°C to +125°C, 1000 cycles
- **Mechanical shock:** 50G half-sine pulse, 11 ms duration
- **Vibration:** Random vibration 5–2000 Hz, 8 hours per axis
- **ESD:** HBM 2 kV, CDM 500 V (higher than consumer grade)
- **EMC:** CISPR 25 radiated and conducted emissions

7.3 Functional Safety (ISO 26262)

Safety-critical sensors (brake pressure, steering angle, airbag) must meet ISO 26262 Automotive Safety Integrity Level (ASIL).

- **ASIL A:** Lowest integrity (e.g., rain sensor)
- **ASIL B:** Medium integrity (e.g., headlight leveling)
- **ASIL C:** High integrity (e.g., lane departure warning)
- **ASIL D:** Highest integrity (e.g., brake pedal position, airbag deployment)

ASIL Requirements:

- Redundant sensors for critical measurements
- Diagnostic coverage > 90% for ASIL D
- Fault-tolerant architecture (e.g., dual-core lockstep)
- Traceability from requirements to test cases

■ **In-House Requirement:** For Tesla and Zoox automotive contracts, verify all sensors are AEC-Q100 qualified and traceable by lot number. Maintain calibration certificates and test records for 15 years per automotive quality standards.

8. IN-HOUSE SENSOR COMPLETE MATRIX

Sensor Type	Examples	Package	Interface	Special Handling	Calibration
Temperature	NTC, RTD, LM75, SHT	0603, SOT-23, QFN	Analog, I2C	Away from heat sources	2-point (0°C, 100°C)
Pressure	BMP280, MPXH, MS5611	LGA, QFN, Metal can	I2C, SPI, Analog	Port orientation, no ultrasonic	Zero + span
Humidity	SHT30, HIH6130, DHT22	SMD, Module	I2C, PWM	Exposed element, no coating	Salt solution 2-point
Proximity	VCNL4040, VL53L0X	SMD, LGA	I2C	Clean optical window	Distance reference
Accelerometer	ADXL345, LIS3DH, MPU6050	LGA, QFN, BGA	I2C, SPI	NO ultrasonic, rigid mount	6-point gravity
Gyroscope	L3GD20, ITG-3200	LGA, QFN	I2C, SPI	NO ultrasonic, axis alignment	Zero-rate + scale
Magnetometer	HMC5883, AK09911	LGA, SMD	I2C, SPI	EMI isolation, shield can	Magnetic reference
Gas	MiCS-5524, MQ-135, CCS811	SMD, Module, TO-5	Analog, I2C, UART	Air access, burn-in 24–48 hr	Known gas concentration
Current	ACS712, INA219, Allegro	SOIC, QFN, SMD	Analog, I2C	High current isolation	Known load
Optical	TCS34725, APDS-9960	SMD, LGA	I2C	Clean lens, ESD protection	Color reference

Table 3: In-House Sensor Complete Matrix

9. QUICK REFERENCE SUMMARY

Sensor Categories:

- **Temperature:** NTC, RTD, digital IC, IR — place away from heat sources
- **Pressure:** MEMS piezoresistive/capacitive — verify port orientation, no ultrasonic
- **Humidity:** Capacitive polymer — exposed element, no coating, gloves handling
- **Proximity/Optical:** IR, ToF, ambient light — clean window, ESD protection
- **Motion:** Accelerometer, gyroscope, IMU — NO ultrasonic, rigid mount, axis alignment
- **Magnetic:** Magnetometer, Hall effect — EMI isolation, shield can
- **Gas:** Electrochemical, MOS, NDIR — air access, burn-in, no encapsulation

MEMS Special Handling:

- **NO ultrasonic cleaning** — spray-in-air or manual wipe only
- **NO mechanical shock** — handle with extreme care
- **MSL 3+ bake required** — 125°C/24 hr if exposed > floor life
- **Vacuum integrity** — do not puncture package cavity
- **Axis orientation** — verify marking before placement

Placement Rules:

- **Temperature:** ≥ 10 mm from heat sources
- **Pressure:** Port faces measurement medium
- **Humidity:** In airflow, not enclosed corners
- **Optical:** Face detection area, no shadowing
- **Motion:** Center of rotation, rigid mounting
- **Magnetic:** Away from inductors and current traces
- **Analog sensors:** Dedicated ground, short shielded traces

Reflow & Cleaning:

- **Standard SAC305:** Peak 245–250°C for most sensors
- **Temperature-sensitive:** Max 230–240°C for some gas/polymer sensors
- **Hand solder:** 315–350°C, 2–5 sec dwell, grounded tools
- **Cleaning:** NO ultrasonic on MEMS; mask optical lenses
- **Dry-out:** 24 hours after cleaning before power-on or calibration

Calibration:

- **Temperature:** 2-point (ice point + boiling point or dry block)
- **Pressure:** Zero + span (vacuum + known pressure)
- **Humidity:** Salt solution 2-point (11% RH + 75% RH)
- **Accelerometer:** 6-point gravity calibration

→ **Gyroscope:** Zero-rate + known rotation rate

Automotive (AEC-Q100):

- **Grade 0:** -40°C to +150°C — under-hood, powertrain
- **Grade 1:** -40°C to +125°C — cabin, body electronics
- **Grade 2:** -40°C to +105°C — infotainment, lighting
- **ISO 26262:** ASIL A to D for functional safety
- **Testing:** HTOL 1000 hr, temp cycling, shock, vibration, ESD, EMC

References & Standards:

- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- IPC-A-610: Acceptability of Electronic Assemblies
- IPC-7093: Design and Assembly Process Implementation for Bottom Termination Components
- AEC-Q100: Failure Mechanism Based Stress Test Qualification for Integrated Circuits
- ISO 26262: Road Vehicles — Functional Safety
- JEDEC J-STD-020: Moisture/Reflow Sensitivity Classification for Nonhermetic Solid State Surface Mount Devices
- JEDEC J-STD-033: Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices
- IPC-1752A: Materials Declaration Management
- Manufacturer Datasheets: Bosch, STMicroelectronics, Analog Devices, Sensirion, NXP, InvenSense/TDK

End of Document — Sensors in PCBA Assembly Complete Guide